

Hrushikesh Etikikota

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Education

Master of Technology - Computer Science and Engineering
Indian Institute of Technology, Gandhinagar

CGPA - 8.71 / 10
July 2022 - June 2024

Experience

LLM - Python Engineer, Turing Enterprises Inc.

Oct 2024 - Present

- Created over 300 notebooks for fine-tuning of LLMs by working across the projects: CoT Reasoning, Function Calling (FC), Agent Completion, Adversarial Self-Correction, Vision FC, and Structured output.
- Promoted to Team Lead for a team of 10 engineers across Agent Completion and Vision-FC projects while consistently reviewing 20 and 50 tasks per day respectively to meet project requirements.
- Developed Python-based test suites for public GitHub repositories on an RLHF project where the LLM models failed to generate correct test cases and evaluate their ability to add new features or functionalities as well.

Teaching Assistant (Part-time) | IIT Gandhinagar

July 2022 – June 2024

- Optimized lab logistics and technical support for 450+ students in Python, Verilog, and Databases, collaborating with a 20-member team to earn faculty recognition for driving exceptional student performance.

Freelance Tutor, Course Hero

June 2021 - July 2022

- Supported over 250 students on their assignments and homework with a 96% helpful rate that enhanced my problem-solving skills on various computer science topics by engaging with a diverse student community.

Projects

A Gaze-Controlled Typing Interface with Context-Based Letter and Word Predictions

- Designed a multi-level tree-based virtual keyboard that supports severe motor and speech-impaired people.
- Integrated statistical letter prediction and word prediction models that help with quick and easy typing.
- Conducted experiments with Mouse and Eye-Tracker and compared the results with the Dasher software.
- Obtained a typing speed of 30.70 and 18.47 letters per minute using the mouse and the eye tracker, respectively, which are nearly 2.7 times more than that of the Dasher with both input devices.

Helmet Detection and License Plate Extraction Using YOLOv3 and Darknet

- Trained a real-time object detection algorithm - YOLOv3 model on 90% of the dataset of 4000 images using the Darknet Framework, achieving an impressive 87.49% average precision on a 10% test set.
- Created a new labelled dataset of 2800 images using Labellmg software and trained the model on 90% of it to achieve a remarkable 97.51% average precision the 10% test set.

Airfoil self-noise Prediction, Breast Cancer Detection, Loan Prediction, Gold Stock Price Prediction, and House Price Prediction

- Implemented and applied the core ML algorithms like Linear Regression, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, SVM, KNN, and XGBoost and identified the best performing model for the given task at hand.

Transformer Architectures for Machine Translation (Spanish to English) and Next Character Prediction

- Build Decoder and Encoder-Decoder Transformers from scratch in PyTorch to master its internal working mechanism.

Technical Skills

- Python ♦ C ♦ C++ ♦ OOPS ♦ SQL ♦ HTML ♦ NumPy ♦ Pandas ♦ scikit-learn ♦ seaborn ♦ Matplotlib ♦ Git ♦ GitHub ♦ Django ♦ Tkinter ♦ PyTorch ♦ TensorFlow ♦ LLM-assisted Development ♦ Deep Learning ♦ Machine Learning ♦ Computer Vision ♦ Prompt Engineering ♦ GenAI ♦ LLM - Supervised Fine-Tuning ♦ LLM - Reinforcement Learning with Human Feedback

Paper Publications

- Published the paper titled “Predictive Tree-based Virtual Keyboard for Improved Gaze Typing” on IEEE SMC2024.

Online Course Certifications

- ‘Getting Started with AWS Machine Learning’ by AWS Specialization.
- ‘Data Science Math Skills’ by Duke University.
- ‘Python for Everybody’ by the University of Michigan.

Achievements

- Received PRATIBHA AWARD for Academic Excellence in the 10th standard issued by Govt. of AP.
- Secured 98.41 percentile (2,981 rank in 187484 students) in EAMCET 2017 (Govt. of AP).
- Attained 98.74 percentile (974 rank in 77257 students) in Graduate Aptitude Test in Engineering (GATE 2022).
- Solved 500+ questions and earned badges on platforms like Hacker Rank, Coding Ninjas, and LeetCode.